

# Meareg Teferi

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GitHub



Portfolio



LinkedIn

**Backend-heavy Software Engineer** specializing in building scalable, highly performant systems using Clean Architecture and Domain-Driven Design. Proficient in Python (FastAPI, Django, Odoo) and JavaScript (Next.js, React). Passionate about advanced system design, asynchronous event-driven microservices, and integrating AI Agents into production workflows. Adept at leveraging proven design patterns to optimize performance and translate complex business rules into maintainable, decoupled code.

## Education

NOVEMBER 2024

**Bachelor of Science in Software Engineering | Mekelle University**

CGPA: 3.81

National Exit Exam: 82%

## Experience

April 2025 – present

**Backend-Heavy Fullstack Engineer | Niyat Consultancy | Addis Ababa, Ethiopia**

- Built scalable, openapi-standard rest APIs utilizing FastAPI, decoupling core business logic through clean architecture principles to ensure long-term maintainability.
- Optimized system performance for high-traffic environments using asynchronous programming and rabbitMQ for reliable, non-blocking event-driven communication.
- Designed and implemented secure authentication systems featuring strict, granular role-based access control (RBAC).
- Developed fault-tolerant third-party payment gateways, ensuring strict state synchronization and automated error recovery for financial transactions.

December 2024 – April 2025

**Backend Engineer | Digital Merkato | Addis Ababa, Ethiopia**

- Developed backend for Ethio Hisab, an accounting SaaS tailored to the Ethiopian market, using Django.
- Applied Domain-Driven Design principles to accurately model local financial and tax regulations into scalable, decoupled backend domains.
- Ensured ACID compliance for financial ledgers while optimizing database queries to handle large-scale financial reporting efficiently.

## Projects

**AI-Powered Project Management System**

FastAPI, Next.js, Redis, PostgreSQL, Pydantic AI

- Architected a high-concurrency task management platform utilizing FastAPI and Redis for sub-second caching and low-latency state management.
- Integrated Pydantic AI Agents to process natural language project descriptions, automatically decomposing unstructured user prompts into structured, actionable task hierarchies and workflows.

### Multi-Branch POS

*FastAPI, React, Tailwind CSS*

- Built a POS system utilizing a heavily optimized FastAPI backend to ensure real-time inventory synchronization across branches.
- Utilized the Repository Pattern and strict Pydantic validation to enforce robust data integrity.

### Custom ERP Solutions

*Odoo, Python, JavaScript, PostgreSQL*

- Customized enterprise resource planning modules across inventory, sales, and manufacturing using Odoo.
- Extended core Python/JavaScript logic to automate complex manufacturing and supply chain workflows.

### Smart Waste Management Platform (Mekelle City Administration)

*Django, React, Redux, Djoser, Material UI, Tailwind CSS*

- Developed a geospatial backend in Django to calculate optimal routing for city-wide waste collection.
- Offloaded heavy background processing and automated report generation to asynchronous message queues.

### Restaurant Management App

*Node.js, React, Redux, MongoDB*

- Engineered a full-stack MERN application featuring end-to-end cart management, secure checkout, order management, and dynamic table reservations governed by centralized Redux state management.

## Skills

**Backend Stacks:** Python, FastAPI, Django, Django REST Framework, Node.js (Express), Odoo, PostgreSQL, MongoDB, Redis, SQLAlchemy, SQLModel, Alembic, Pydantic AI, LangGraph.

**Frontend Stacks:** JavaScript, React.js, Next.js, Redux, Tailwind CSS, Material UI.

**Architectures:** Clean Architecture, Domain-Driven Design (DDD), Microservices, Event-Driven Architecture, RESTful API Architecture.

**Patterns:** Repository Pattern, Publisher/Subscriber, Asynchronous Programming (AsyncIO), Role-Based Access Control, Model-Controller-View, JWT (Stateless Authentication).

**Tools:** Docker, Docker Compose, RabbitMQ, Traefik (Reverse Proxy), Nginx, Git/GitHub Actions (CI/CD), Pytest, ruff, mypy, Playwright, OpenAPI/Swagger, Postman.